Intro to AI – Practical Assignment 2

How to run

Run ***\project\src\Env\Main.java*** and provide it with the path to the input file as argument the first argument.

Project Structure

- **\src\Env** contain the main setup, environment and Enums
- **\src\Graph** contain the Graph, State, Simulator
- **\src\Player** contain an abstract Player class and the setup for the loyalty game
- **\src\Test** contain Javadocs of the input file and test generator

A description of your heuristic evaluation function

The heuristic evaluation function I chose is “A Bird in the Hand is Worth Two in the Bush”.

Meaning that a delivered package is worth double points of a package which was picked up but wasn’t delivered yet.

Player evaluation:

DELIVERED_PACKAGES * 2 + PICKED_UP_PACKAGES * 1

I’ve chose this heuristic because it fits the symmetry where if a player picks up a package, it is worth double than the actual score for delivering said package for him.

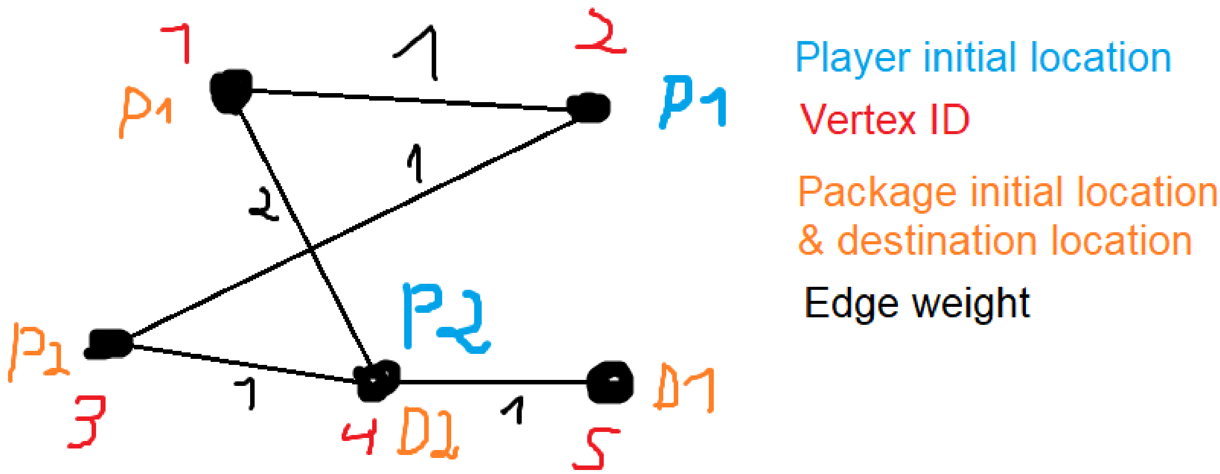
Because he can deliver it himself, and also prevent the other player from taking it.

A differentiating example

For all 3 loyalty options:

1. Adversarial (zero sum game): each agent aims to maximize its own individual score (number of packages deliverd on time) minus the opposing agent's score. That is, TS1=IS1-IS2 and TS2=IS2-IS1. Here you should implement an "optimal" agent, using mini-max, with alpha-beta pruning.
2. A semi-cooperative game: each agent tries to maximize its own individual score. The agent disregards the other agent score, except that ties are broken **cooperatively**. THat is, TS1=IS1, breaking ties in favor of greater IS2.
3. A **fully cooperative** game: both agents aim to maximize the sum of individual scores, so TS1=TS2=IS1+IS2.

Graph Image:



Example input code:

```
#N 5
#E1 1 2 W1
#E2 2 3 W1
#E3 3 4 W1
#E4 1 4 W2
#E5 4 5 W1

#P1 V1 0 D V5 6
#P2 V3 0 D V4 3

#M V2 V4 <M1 / M2 / M3> 5
```

Output:

Note: for convenience, if an agent is on its way to vertex x, it is marked as his location until he gets there.

(Xi, Yi) mean agent #1's location is X, and agent #2's location is Y at time i.

- Mode 1 – Adversarial
  [
  (V2, V4),
  (V2, V1),
  (V1, V1),
  (V1, V1),
  (V2, V2),
  (V3, V1)
  ]

- Mode 2 – Semi cooperative
  [
  (V2, V4),
  (V3, V1),
  (V4, V1),
  (V1, V4),
  (V1, V4),
  (V2, V5)
  ]

- Mode 3 – Fully cooperative

[

(V2, V4),

(V1, V3),

(V4, V4),

(V4, V1),

(V5,V1)
]

Explanation:

- On game mode 3 (Fully cooperative), both players are working together to get the highest combined score as fast as possible. Resulting in a game of 5 moves.

- On game mode 2 (Semi cooperative), both players pick up and deliver a single package each, and because they are aiming to first maximize their own score, they take a longer detour until the packages are delivered, resulting in a game of 6 moves.

- On game mode 1 (Adversarial) , both players are aiming to maximize the difference between their score and their opponent's.
  Since both players are calculating that the minimax value for both is 0 whether they decide to pick up the packages or not, and they both know that the opponent knows the same and is also deterministic.
  They decide to not pick up any package - resulting in a mutual score of 0.

Full game logs:

- Adversarial:

```
=============================================
Global time: 0

Player #1 Loyalty Player in  Location: V2 Score: 0
Player #2 Loyalty Player in  Location: V4 Score: 0

=============================================
Global time: 1


Player #1 Loyalty Player in  Location: V2 Score: 0
Player #2 Loyalty Player in 2 units of time will arive to  Location: V1 Score: 0

=============================================
Global time: 2


Player #1 Loyalty Player in  Location: V1 Score: 0
Player #2 Loyalty Player in  Location: V1 Score: 0

=============================================
Global time: 3


Player #1 Loyalty Player in  Location: V1 Score: 0
Player #2 Loyalty Player in  Location: V1 Score: 0

=============================================
Global time: 4


Player #1 Loyalty Player in  Location: V2 Score: 0
Player #2 Loyalty Player in  Location: V2 Score: 0

=============================================
Global time: 5

Player #1 Loyalty Player in  Location: V3 Score: 0
Player #2 Loyalty Player in  Location: V1 Score: 0

The game is over!
**********   Score table:    **********
Player 0 of type Loyalty Player with score: 0
Player 1 of type Loyalty Player with score: 0
```

- Semi-Cooperative:

===

Global time: 0

Player #1 Loyalty Player in Location: V2 Score: 0

Player #2 Loyalty Player in Location: V4 Score: 0

===

Global time: 1

Player #1 Loyalty Player in Location: V3 Score: 0

Player #2 Loyalty Player in 2 units of time will arive to Location: V1 Score: 0

===

Global time: 2

Player #1 Loyalty Player in Location: V4 Score: 0

Player #2 Loyalty Player in Location: V1 Score: 0

===

Global time: 3

Player #1 Loyalty Player in 2 units of time will arrive to Location: V1 Score: 1

Player #2 Loyalty Player in 2 units of time will arrive to Location: V4 Score: 0

===

Global time: 4

Player #1 Loyalty Player in Location: V1 Score: 1

Player #2 Loyalty Player in Location: V4 Score: 0

===

Global time: 5

Player #1 Loyalty Player in Location: V2 Score: 1

Player #2 Loyalty Player in Location: V5 Score: 0

The game is over!

**********   Score table: **********

Player 0 of type Loyalty Player with score: 1

Player 1 of type Loyalty Player with score: 1

- Fullly-Cooperative

==

Global time: 0

Player #1 Loyalty Player in Location: V2 Score: 0

Player #2 Loyalty Player in Location: V4 Score: 0

==

Global time: 1

Player #1 Loyalty Player in Location: V1 Score: 0

Player #2 Loyalty Player in Location: V3 Score: 0

==

Global time: 2

Player #1 Loyalty Player in 2 units of time will arive to Location: V4 Score: 0

Player #2 Loyalty Player in Location: V4 Score: 0

==

Global time: 3

Player #1 Loyalty Player in Location: V4 Score: 0

Player #2 Loyalty Player in 2 units of time will arive to Location: V1 Score: 1

==

Global time: 4

Player #1 Loyalty Player in Location: V5 Score: 0

Player #2 Loyalty Player in Location: V1 Score: 1

The game is over!

**********   Score table: **********

Player 0 of type Loyalty Player with score: 1

Player 1 of type Loyalty Player with score: 1